

Enterprise Infrastructure Knowledge Base

Complete Incident Response, Telemetry Routing, & Operational Diagnostics Guide

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Target Systems: API Gateways, PostgreSQL, Redis, Auth Services, Event Streams

Generated For: Agentic Support & RAG Vector Pipeline

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KB-101: Troubleshooting API Gateway 504 Timeouts

System Subservice: Gateway | **Category:** API Error | **Severity Level:** High

1. Diagnostic Symptoms

Clients or automated monitors report standard failure patterns under `/v1/services/gateway`. Alerts fire in Prometheus/Grafana when error rates exceed threshold parameters over a 5-minute rolling window.

2. Root Cause Analysis

Upstream failure, resource depletion, network packet loss, or misconfigured quota policies. Requires validation against live Prometheus metrics before applying manual intervention or triggering automated failover routines.

3. Step-by-Step Resolution Workflow

- 1. Query Live Telemetry:** Check Prometheus/Grafana endpoints to confirm if the anomaly is regional or isolated to a specific tenant.
- 2. Evaluate Active Incidents:** Cross-reference active outage logs (e.g., Datadog/Grafana incident IDs) to prevent unnecessary user-level troubleshooting during system-wide degradation.
- 3. Perform Remediations:**
 - Execute graceful process restarts or scaling operations: `kubectl rollout restart deployment/gateway`
 - Flush ephemeral memory or adjust client connection parameters if limits were breached.
 - Notify stakeholders if recovery times exceed standard Service Level Objectives (SLOs).

4. RAG Retrieval Metadata

Vector ID: doc_kb_101

Recommended Action Tag: api_error_mitigation_v2

ChromaDB Embedding Filter: `{"category": "API Error", "service": "Gateway"}`

KB-102: Resolving Postgres Connection Pool Exhaustion

System Subservice: Database | **Category:** Database | **Severity Level:** High

1. Diagnostic Symptoms

Clients or automated monitors report standard failure patterns under `/v1/services/database`. Alerts fire in Prometheus/Grafana when error rates exceed threshold parameters over a 5-minute rolling window.

2. Root Cause Analysis

Upstream failure, resource depletion, network packet loss, or misconfigured quota policies. Requires validation against live Prometheus metrics before applying manual intervention or triggering automated failover routines.

3. Step-by-Step Resolution Workflow

- 1. Query Live Telemetry:** Check Prometheus/Grafana endpoints to confirm if the anomaly is regional or isolated to a specific tenant.
- 2. Evaluate Active Incidents:** Cross-reference active outage logs (e.g., Datadog/Grafana incident IDs) to prevent unnecessary user-level troubleshooting during system-wide degradation.
- 3. Perform Remediations:**
 - Execute graceful process restarts or scaling operations: `kubectl rollout restart deployment/database`
 - Flush ephemeral memory or adjust client connection parameters if limits were breached.
 - Notify stakeholders if recovery times exceed standard Service Level Objectives (SLOs).

4. RAG Retrieval Metadata

Vector ID: doc_kb_102

Recommended Action Tag: database_mitigation_v2

ChromaDB Embedding Filter: `{"category": "Database", "service": "Database"}`

KB-103: Handling HTTP 429 Rate Limit Exceeded

System Subservice: Rate Limiter | **Category:** API Error | **Severity Level:** Medium

1. Diagnostic Symptoms

Clients or automated monitors report standard failure patterns under `/v1/services/rate-limiter`. Alerts fire in Prometheus/Grafana when error rates exceed threshold parameters over a 5-minute rolling window.

2. Root Cause Analysis

Upstream failure, resource depletion, network packet loss, or misconfigured quota policies. Requires validation against live Prometheus metrics before applying manual intervention or triggering automated failover routines.

3. Step-by-Step Resolution Workflow

- 1. Query Live Telemetry:** Check Prometheus/Grafana endpoints to confirm if the anomaly is regional or isolated to a specific tenant.
- 2. Evaluate Active Incidents:** Cross-reference active outage logs (e.g., Datadog/Grafana incident IDs) to prevent unnecessary user-level troubleshooting during system-wide degradation.
- 3. Perform Remediations:**
 - Execute graceful process restarts or scaling operations: `kubectl rollout restart deployment/rate-limiter`
 - Flush ephemeral memory or adjust client connection parameters if limits were breached.
 - Notify stakeholders if recovery times exceed standard Service Level Objectives (SLOs).

4. RAG Retrieval Metadata

Vector ID: doc_kb_103

Recommended Action Tag: api_error_mitigation_v2

ChromaDB Embedding Filter: `{"category": "API Error", "service": "Rate Limiter"}`

KB-104: Fixing 401 Unauthorized and Expired JWT Tokens

System Subservice: Auth Gateway | **Category:** Authentication | **Severity Level:** Medium

1. Diagnostic Symptoms

Clients or automated monitors report standard failure patterns under `/v1/services/auth-gateway`. Alerts fire in Prometheus/Grafana when error rates exceed threshold parameters over a 5-minute rolling window.

2. Root Cause Analysis

Upstream failure, resource depletion, network packet loss, or misconfigured quota policies. Requires validation against live Prometheus metrics before applying manual intervention or triggering automated failover routines.

3. Step-by-Step Resolution Workflow

- 1. Query Live Telemetry:** Check Prometheus/Grafana endpoints to confirm if the anomaly is regional or isolated to a specific tenant.
- 2. Evaluate Active Incidents:** Cross-reference active outage logs (e.g., Datadog/Grafana incident IDs) to prevent unnecessary user-level troubleshooting during system-wide degradation.
- 3. Perform Remediations:**
 - Execute graceful process restarts or scaling operations: `kubectl rollout restart deployment/auth-gateway`
 - Flush ephemeral memory or adjust client connection parameters if limits were breached.
 - Notify stakeholders if recovery times exceed standard Service Level Objectives (SLOs).

4. RAG Retrieval Metadata

Vector ID: doc_kb_104

Recommended Action Tag: authentication_mitigation_v2

ChromaDB Embedding Filter: `{"category": "Authentication", "service": "Auth Gateway"}`

KB-105: Webhook Delivery Failure & Retry Policies

System Subservice: Event Bus | **Category:** Infrastructure | **Severity Level:** Medium

1. Diagnostic Symptoms

Clients or automated monitors report standard failure patterns under `/v1/services/event-bus`. Alerts fire in Prometheus/Grafana when error rates exceed threshold parameters over a 5-minute rolling window.

2. Root Cause Analysis

Upstream failure, resource depletion, network packet loss, or misconfigured quota policies. Requires validation against live Prometheus metrics before applying manual intervention or triggering automated failover routines.

3. Step-by-Step Resolution Workflow

- 1. Query Live Telemetry:** Check Prometheus/Grafana endpoints to confirm if the anomaly is regional or isolated to a specific tenant.
- 2. Evaluate Active Incidents:** Cross-reference active outage logs (e.g., Datadog/Grafana incident IDs) to prevent unnecessary user-level troubleshooting during system-wide degradation.
- 3. Perform Remediations:**
 - Execute graceful process restarts or scaling operations: `kubectl rollout restart deployment/event-bus`
 - Flush ephemeral memory or adjust client connection parameters if limits were breached.
 - Notify stakeholders if recovery times exceed standard Service Level Objectives (SLOs).

4. RAG Retrieval Metadata

Vector ID: doc_kb_105

Recommended Action Tag: infrastructure_mitigation_v2

ChromaDB Embedding Filter: `{"category": "Infrastructure", "service": "Event Bus"}`

KB-106: Resolving Cross-Origin Resource Sharing (CORS) Errors

System Subservice: Gateway | **Category:** API Error | **Severity Level:** Low

1. Diagnostic Symptoms

Clients or automated monitors report standard failure patterns under `/v1/services/gateway`. Alerts fire in Prometheus/Grafana when error rates exceed threshold parameters over a 5-minute rolling window.

2. Root Cause Analysis

Upstream failure, resource depletion, network packet loss, or misconfigured quota policies. Requires validation against live Prometheus metrics before applying manual intervention or triggering automated failover routines.

3. Step-by-Step Resolution Workflow

- 1. Query Live Telemetry:** Check Prometheus/Grafana endpoints to confirm if the anomaly is regional or isolated to a specific tenant.
- 2. Evaluate Active Incidents:** Cross-reference active outage logs (e.g., Datadog/Grafana incident IDs) to prevent unnecessary user-level troubleshooting during system-wide degradation.
- 3. Perform Remediations:**
 - Execute graceful process restarts or scaling operations: `kubectl rollout restart deployment/gateway`
 - Flush ephemeral memory or adjust client connection parameters if limits were breached.
 - Notify stakeholders if recovery times exceed standard Service Level Objectives (SLOs).

4. RAG Retrieval Metadata

Vector ID: doc_kb_106

Recommended Action Tag: api_error_mitigation_v2

ChromaDB Embedding Filter: `{"category": "API Error", "service": "Gateway"}`

KB-107: Managing Redis Out-Of-Memory (OOM) Evictions

System Subservice: Cache Cluster | **Category:** Database | **Severity Level:** High

1. Diagnostic Symptoms

Clients or automated monitors report standard failure patterns under `/v1/services/cache-cluster`. Alerts fire in Prometheus/Grafana when error rates exceed threshold parameters over a 5-minute rolling window.

2. Root Cause Analysis

Upstream failure, resource depletion, network packet loss, or misconfigured quota policies. Requires validation against live Prometheus metrics before applying manual intervention or triggering automated failover routines.

3. Step-by-Step Resolution Workflow

- 1. Query Live Telemetry:** Check Prometheus/Grafana endpoints to confirm if the anomaly is regional or isolated to a specific tenant.
- 2. Evaluate Active Incidents:** Cross-reference active outage logs (e.g., Datadog/Grafana incident IDs) to prevent unnecessary user-level troubleshooting during system-wide degradation.
- 3. Perform Remediations:**
 - Execute graceful process restarts or scaling operations: `kubectl rollout restart deployment/cache-cluster`
 - Flush ephemeral memory or adjust client connection parameters if limits were breached.
 - Notify stakeholders if recovery times exceed standard Service Level Objectives (SLOs).

4. RAG Retrieval Metadata

Vector ID: doc_kb_107

Recommended Action Tag: database_mitigation_v2

ChromaDB Embedding Filter: `{"category": "Database", "service": "Cache Cluster"}`

KB-108: Edge Proxy SSL / TLS Handshake Failures

System Subservice: TLS/Edge | **Category:** Infrastructure | **Severity Level:** High

1. Diagnostic Symptoms

Clients or automated monitors report standard failure patterns under `/v1/services/tls/edge`. Alerts fire in Prometheus/Grafana when error rates exceed threshold parameters over a 5-minute rolling window.

2. Root Cause Analysis

Upstream failure, resource depletion, network packet loss, or misconfigured quota policies. Requires validation against live Prometheus metrics before applying manual intervention or triggering automated failover routines.

3. Step-by-Step Resolution Workflow

- 1. Query Live Telemetry:** Check Prometheus/Grafana endpoints to confirm if the anomaly is regional or isolated to a specific tenant.
- 2. Evaluate Active Incidents:** Cross-reference active outage logs (e.g., Datadog/Grafana incident IDs) to prevent unnecessary user-level troubleshooting during system-wide degradation.
- 3. Perform Remediations:**
 - Execute graceful process restarts or scaling operations: `kubectl rollout restart deployment/tls/edge`
 - Flush ephemeral memory or adjust client connection parameters if limits were breached.
 - Notify stakeholders if recovery times exceed standard Service Level Objectives (SLOs).

4. RAG Retrieval Metadata

Vector ID: doc_kb_108

Recommended Action Tag: infrastructure_mitigation_v2

ChromaDB Embedding Filter: `{"category": "Infrastructure", "service": "TLS/Edge"}`

KB-109: 402 Payment Required & Account Suspension

System Subservice: Account Portal | **Category:** Billing | **Severity Level:** High

1. Diagnostic Symptoms

Clients or automated monitors report standard failure patterns under `/v1/services/account-portal`. Alerts fire in Prometheus/Grafana when error rates exceed threshold parameters over a 5-minute rolling window.

2. Root Cause Analysis

Upstream failure, resource depletion, network packet loss, or misconfigured quota policies. Requires validation against live Prometheus metrics before applying manual intervention or triggering automated failover routines.

3. Step-by-Step Resolution Workflow

- 1. Query Live Telemetry:** Check Prometheus/Grafana endpoints to confirm if the anomaly is regional or isolated to a specific tenant.
- 2. Evaluate Active Incidents:** Cross-reference active outage logs (e.g., Datadog/Grafana incident IDs) to prevent unnecessary user-level troubleshooting during system-wide degradation.
- 3. Perform Remediations:**
 - Execute graceful process restarts or scaling operations: `kubectl rollout restart deployment/account-portal`
 - Flush ephemeral memory or adjust client connection parameters if limits were breached.
 - Notify stakeholders if recovery times exceed standard Service Level Objectives (SLOs).

4. RAG Retrieval Metadata

Vector ID: doc_kb_109

Recommended Action Tag: billing_mitigation_v2

ChromaDB Embedding Filter: `{"category": "Billing", "service": "Account Portal"}`

KB-110: Presigned S3 Image & Document Upload Failures

System Subservice: Storage | **Category:** Infrastructure | **Severity Level:** Medium

1. Diagnostic Symptoms

Clients or automated monitors report standard failure patterns under `/v1/services/storage`. Alerts fire in Prometheus/Grafana when error rates exceed threshold parameters over a 5-minute rolling window.

2. Root Cause Analysis

Upstream failure, resource depletion, network packet loss, or misconfigured quota policies. Requires validation against live Prometheus metrics before applying manual intervention or triggering automated failover routines.

3. Step-by-Step Resolution Workflow

- 1. Query Live Telemetry:** Check Prometheus/Grafana endpoints to confirm if the anomaly is regional or isolated to a specific tenant.
- 2. Evaluate Active Incidents:** Cross-reference active outage logs (e.g., Datadog/Grafana incident IDs) to prevent unnecessary user-level troubleshooting during system-wide degradation.
- 3. Perform Remediations:**
 - Execute graceful process restarts or scaling operations: `kubectl rollout restart deployment/storage`
 - Flush ephemeral memory or adjust client connection parameters if limits were breached.
 - Notify stakeholders if recovery times exceed standard Service Level Objectives (SLOs).

4. RAG Retrieval Metadata

Vector ID: doc_kb_110

Recommended Action Tag: infrastructure_mitigation_v2

ChromaDB Embedding Filter: `{"category": "Infrastructure", "service": "Storage"}`

KB-111: Kafka Consumer Group Lag and Rebalance Storms

System Subservice: Kafka Streaming | **Category:** Infrastructure | **Severity Level:** High

1. Diagnostic Symptoms

Clients or automated monitors report standard failure patterns under `/v1/services/kafka-streaming`. Alerts fire in Prometheus/Grafana when error rates exceed threshold parameters over a 5-minute rolling window.

2. Root Cause Analysis

Upstream failure, resource depletion, network packet loss, or misconfigured quota policies. Requires validation against live Prometheus metrics before applying manual intervention or triggering automated failover routines.

3. Step-by-Step Resolution Workflow

- 1. Query Live Telemetry:** Check Prometheus/Grafana endpoints to confirm if the anomaly is regional or isolated to a specific tenant.
- 2. Evaluate Active Incidents:** Cross-reference active outage logs (e.g., Datadog/Grafana incident IDs) to prevent unnecessary user-level troubleshooting during system-wide degradation.
- 3. Perform Remediations:**
 - Execute graceful process restarts or scaling operations: `kubectl rollout restart deployment/kafka-streaming`
 - Flush ephemeral memory or adjust client connection parameters if limits were breached.
 - Notify stakeholders if recovery times exceed standard Service Level Objectives (SLOs).

4. RAG Retrieval Metadata

Vector ID: doc_kb_111

Recommended Action Tag: infrastructure_mitigation_v2

ChromaDB Embedding Filter: `{"category": "Infrastructure", "service": "Kafka Streaming"}`

KB-112: Elasticsearch Unassigned Shards and Red Cluster State

System Subservice: Search Engine | **Category:** Database | **Severity Level:** High

1. Diagnostic Symptoms

Clients or automated monitors report standard failure patterns under `/v1/services/search-engine`. Alerts fire in Prometheus/Grafana when error rates exceed threshold parameters over a 5-minute rolling window.

2. Root Cause Analysis

Upstream failure, resource depletion, network packet loss, or misconfigured quota policies. Requires validation against live Prometheus metrics before applying manual intervention or triggering automated failover routines.

3. Step-by-Step Resolution Workflow

- 1. Query Live Telemetry:** Check Prometheus/Grafana endpoints to confirm if the anomaly is regional or isolated to a specific tenant.
- 2. Evaluate Active Incidents:** Cross-reference active outage logs (e.g., Datadog/Grafana incident IDs) to prevent unnecessary user-level troubleshooting during system-wide degradation.
- 3. Perform Remediations:**
 - Execute graceful process restarts or scaling operations: `kubectl rollout restart deployment/search-engine`
 - Flush ephemeral memory or adjust client connection parameters if limits were breached.
 - Notify stakeholders if recovery times exceed standard Service Level Objectives (SLOs).

4. RAG Retrieval Metadata

Vector ID: doc_kb_112

Recommended Action Tag: database_mitigation_v2

ChromaDB Embedding Filter: `{"category": "Database", "service": "Search Engine"}`

KB-113: gRPC RPC Deadline Exceeded and Transport Breached

System Subservice: Microservices Router | **Category:** API Error | **Severity Level:** High

1. Diagnostic Symptoms

Clients or automated monitors report standard failure patterns under `/v1/services/microservices-router`. Alerts fire in Prometheus/Grafana when error rates exceed threshold parameters over a 5-minute rolling window.

2. Root Cause Analysis

Upstream failure, resource depletion, network packet loss, or misconfigured quota policies. Requires validation against live Prometheus metrics before applying manual intervention or triggering automated failover routines.

3. Step-by-Step Resolution Workflow

- 1. Query Live Telemetry:** Check Prometheus/Grafana endpoints to confirm if the anomaly is regional or isolated to a specific tenant.
- 2. Evaluate Active Incidents:** Cross-reference active outage logs (e.g., Datadog/Grafana incident IDs) to prevent unnecessary user-level troubleshooting during system-wide degradation.
- 3. Perform Remediations:**
 - Execute graceful process restarts or scaling operations: `kubectl rollout restart deployment/microservices-router`
 - Flush ephemeral memory or adjust client connection parameters if limits were breached.
 - Notify stakeholders if recovery times exceed standard Service Level Objectives (SLOs).

4. RAG Retrieval Metadata

Vector ID: doc_kb_113

Recommended Action Tag: api_error_mitigation_v2

ChromaDB Embedding Filter: `{"category": "API Error", "service": "Microservices Router"}`

KB-114: DNS Resolution Failures and Transitory NXDOMAIN Errors

System Subservice: Core Networking | **Category:** Infrastructure | **Severity Level:** High

1. Diagnostic Symptoms

Clients or automated monitors report standard failure patterns under `/v1/services/core-networking`. Alerts fire in Prometheus/Grafana when error rates exceed threshold parameters over a 5-minute rolling window.

2. Root Cause Analysis

Upstream failure, resource depletion, network packet loss, or misconfigured quota policies. Requires validation against live Prometheus metrics before applying manual intervention or triggering automated failover routines.

3. Step-by-Step Resolution Workflow

- 1. Query Live Telemetry:** Check Prometheus/Grafana endpoints to confirm if the anomaly is regional or isolated to a specific tenant.
- 2. Evaluate Active Incidents:** Cross-reference active outage logs (e.g., Datadog/Grafana incident IDs) to prevent unnecessary user-level troubleshooting during system-wide degradation.
- 3. Perform Remediations:**
 - Execute graceful process restarts or scaling operations: `kubectl rollout restart deployment/core-networking`
 - Flush ephemeral memory or adjust client connection parameters if limits were breached.
 - Notify stakeholders if recovery times exceed standard Service Level Objectives (SLOs).

4. RAG Retrieval Metadata

Vector ID: doc_kb_114

Recommended Action Tag: infrastructure_mitigation_v2

ChromaDB Embedding Filter: `{"category": "Infrastructure", "service": "Core Networking"}`

KB-115: Kubernetes Pod OOMKilled (Exit Code 137)

Recovery

System Subservice: K8s Worker Nodes | **Category:** Infrastructure | **Severity Level:** High

1. Diagnostic Symptoms

Clients or automated monitors report standard failure patterns under `/v1/services/k8s-worker-nodes`. Alerts fire in Prometheus/Grafana when error rates exceed threshold parameters over a 5-minute rolling window.

2. Root Cause Analysis

Upstream failure, resource depletion, network packet loss, or misconfigured quota policies. Requires validation against live Prometheus metrics before applying manual intervention or triggering automated failover routines.

3. Step-by-Step Resolution Workflow

- 1. Query Live Telemetry:** Check Prometheus/Grafana endpoints to confirm if the anomaly is regional or isolated to a specific tenant.
- 2. Evaluate Active Incidents:** Cross-reference active outage logs (e.g., Datadog/Grafana incident IDs) to prevent unnecessary user-level troubleshooting during system-wide degradation.
- 3. Perform Remediations:**
 - Execute graceful process restarts or scaling operations: `kubectl rollout restart deployment/k8s-worker-nodes`
 - Flush ephemeral memory or adjust client connection parameters if limits were breached.
 - Notify stakeholders if recovery times exceed standard Service Level Objectives (SLOs).

4. RAG Retrieval Metadata

Vector ID: doc_kb_115

Recommended Action Tag: infrastructure_mitigation_v2

ChromaDB Embedding Filter: `{"category": "Infrastructure", "service": "K8s Worker Nodes"}`

KB-116: GraphQL Query Complexity Limits and Depth Truncation

System Subservice: GraphQL API Gateway | **Category:** API Error | **Severity Level:** Medium

1. Diagnostic Symptoms

Clients or automated monitors report standard failure patterns under `/v1/services/graphql-api-gateway`. Alerts fire in Prometheus/Grafana when error rates exceed threshold parameters over a 5-minute rolling window.

2. Root Cause Analysis

Upstream failure, resource depletion, network packet loss, or misconfigured quota policies. Requires validation against live Prometheus metrics before applying manual intervention or triggering automated failover routines.

3. Step-by-Step Resolution Workflow

- 1. Query Live Telemetry:** Check Prometheus/Grafana endpoints to confirm if the anomaly is regional or isolated to a specific tenant.
- 2. Evaluate Active Incidents:** Cross-reference active outage logs (e.g., Datadog/Grafana incident IDs) to prevent unnecessary user-level troubleshooting during system-wide degradation.
- 3. Perform Remediations:**
 - Execute graceful process restarts or scaling operations: `kubectl rollout restart deployment/graphql-api-gateway`
 - Flush ephemeral memory or adjust client connection parameters if limits were breached.
 - Notify stakeholders if recovery times exceed standard Service Level Objectives (SLOs).

4. RAG Retrieval Metadata

Vector ID: doc_kb_116

Recommended Action Tag: api_error_mitigation_v2

ChromaDB Embedding Filter: `{"category": "API Error", "service": "GraphQL API Gateway"}`

KB-117: RabbitMQ Queue Memory Alarms and Producer Blocking

System Subservice: Message Queue | **Category:** Infrastructure | **Severity Level:** High

1. Diagnostic Symptoms

Clients or automated monitors report standard failure patterns under `/v1/services/message-queue`. Alerts fire in Prometheus/Grafana when error rates exceed threshold parameters over a 5-minute rolling window.

2. Root Cause Analysis

Upstream failure, resource depletion, network packet loss, or misconfigured quota policies. Requires validation against live Prometheus metrics before applying manual intervention or triggering automated failover routines.

3. Step-by-Step Resolution Workflow

- 1. Query Live Telemetry:** Check Prometheus/Grafana endpoints to confirm if the anomaly is regional or isolated to a specific tenant.
- 2. Evaluate Active Incidents:** Cross-reference active outage logs (e.g., Datadog/Grafana incident IDs) to prevent unnecessary user-level troubleshooting during system-wide degradation.
- 3. Perform Remediations:**
 - Execute graceful process restarts or scaling operations: `kubectl rollout restart deployment/message-queue`
 - Flush ephemeral memory or adjust client connection parameters if limits were breached.
 - Notify stakeholders if recovery times exceed standard Service Level Objectives (SLOs).

4. RAG Retrieval Metadata

Vector ID: doc_kb_117

Recommended Action Tag: infrastructure_mitigation_v2

ChromaDB Embedding Filter: `{"category": "Infrastructure", "service": "Message Queue"}`

KB-118: Vault Secret Renewal Expiration and Token Lease Loss

System Subservice: Secrets Manager | **Category:** Security | **Severity Level:** High

1. Diagnostic Symptoms

Clients or automated monitors report standard failure patterns under `/v1/services/secrets-manager`. Alerts fire in Prometheus/Grafana when error rates exceed threshold parameters over a 5-minute rolling window.

2. Root Cause Analysis

Upstream failure, resource depletion, network packet loss, or misconfigured quota policies. Requires validation against live Prometheus metrics before applying manual intervention or triggering automated failover routines.

3. Step-by-Step Resolution Workflow

- 1. Query Live Telemetry:** Check Prometheus/Grafana endpoints to confirm if the anomaly is regional or isolated to a specific tenant.
- 2. Evaluate Active Incidents:** Cross-reference active outage logs (e.g., Datadog/Grafana incident IDs) to prevent unnecessary user-level troubleshooting during system-wide degradation.
- 3. Perform Remediations:**
 - Execute graceful process restarts or scaling operations: `kubectl rollout restart deployment/secrets-manager`
 - Flush ephemeral memory or adjust client connection parameters if limits were breached.
 - Notify stakeholders if recovery times exceed standard Service Level Objectives (SLOs).

4. RAG Retrieval Metadata

Vector ID: doc_kb_118

Recommended Action Tag: security_mitigation_v2

ChromaDB Embedding Filter: `{"category": "Security", "service": "Secrets Manager"}`

KB-119: CDN Cache Invalidation Delays and Stale Asset Serving

System Subservice: Edge CDN | **Category:** Infrastructure | **Severity Level:** Low

1. Diagnostic Symptoms

Clients or automated monitors report standard failure patterns under `/v1/services/edge-cdn`. Alerts fire in Prometheus/Grafana when error rates exceed threshold parameters over a 5-minute rolling window.

2. Root Cause Analysis

Upstream failure, resource depletion, network packet loss, or misconfigured quota policies. Requires validation against live Prometheus metrics before applying manual intervention or triggering automated failover routines.

3. Step-by-Step Resolution Workflow

- 1. Query Live Telemetry:** Check Prometheus/Grafana endpoints to confirm if the anomaly is regional or isolated to a specific tenant.
- 2. Evaluate Active Incidents:** Cross-reference active outage logs (e.g., Datadog/Grafana incident IDs) to prevent unnecessary user-level troubleshooting during system-wide degradation.
- 3. Perform Remediations:**
 - Execute graceful process restarts or scaling operations: `kubectl rollout restart deployment/edge-cdn`
 - Flush ephemeral memory or adjust client connection parameters if limits were breached.
 - Notify stakeholders if recovery times exceed standard Service Level Objectives (SLOs).

4. RAG Retrieval Metadata

Vector ID: doc_kb_119

Recommended Action Tag: infrastructure_mitigation_v2

ChromaDB Embedding Filter: `{"category": "Infrastructure", "service": "Edge CDN"}`